

SATGUN SINGH SODHI

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EDUCATION

B.Tech Computer Science and Engineering Vellore Institute of Technology 9.38 CGPA	Vellore 2023-Present
Class XII Delhi World Public School 91.8%	Nerul 2023
Class X Apeejay School 90.6%	Nerul 2021

EXPERIENCE

Machine Learning Intern

Reliance Industries Limited | May 2025 - July 2025, RCP, Ghansoli, Mumbai

- Engineered 4+ domain-specific features (e.g., normalized flow deltas, cosine similarity) to accurately capture physical flow behavior, improving anomaly detection precision by 30%.
- Mapped real-world pipeline behavior into mathematical patterns, reducing false positives by 25% through custom thresholding and directional flow logic. Identified system-specific deviations (e.g., obstruction/backflow) and designed features that replicated these behaviors in model inputs, improving interpretability and reliability.
- Collaborated with field teams to validate model assumptions against actual process behavior, ensuring real-world alignment and boosting stakeholder trust.

PROJECTS

StreamFlix – Netflix Clone with Watch Party Feature

Course Project • GitHub: github.com/satgunsodhi/StreamFlix | Live Demo: stream-watch-party.onrender.com

- Architected modular content browsing and playback components using React 19, Zustand, and Chakra UI, enabling **responsive dark/light theme switching**.
- Implemented real-time bi-directional communication with WebSockets, achieving seamless video synchronization for **multi-user watch parties**.
- Designed a **responsive UI** adaptable across desktop and mobile, improving user experience and engagement.
- Collaborated in a team of 3** to deliver full-stack features including content browsing, playback controls, and user sessions under course evaluation.

Pokemon Color Classifier – Machine Learning Image Classification System

Personal Project • GitHub: github.com/satgunsodhi/Image-Color-Classifer

- Built a hybrid pipeline to **classify 800+ Pokémon images into 10 color classes** using K-Means, Random Forest, SVM, and CNNs.
- Extracted dominant color features via K-Means clustering, achieving up to **87% accuracy** with traditional ML methods.
- Designed and trained a Convolutional Neural Network (CNN)** leveraging spatial and color distribution features for **improved accuracy**.
- Applied OpenCV and Scikit-learn** for preprocessing, contour extraction, and evaluation on noisy real-world datasets.

Forgery Classifier – Signature Forgery Detection System

Personal Project • GitHub: github.com/satgunsodhi/Forgery-Classifer

- Implemented and benchmarked 4 ML models** (SVC, KNN, Logistic Regression, GaussianNB), achieving up to 94% accuracy on clean signature datasets.
- Applied Neural Style Transfer (NST)** to simulate adversarial forgeries, exposing 30–50% accuracy drops and highlighting model fragility.
- Built a signature preprocessing pipeline** with OpenCV for contour extraction, noise reduction, and stroke enhancement.
- Designed a **comparative evaluation framework** to test robustness of ML models under adversarial transformations.

SKILLS & ABILITIES

Technical Skills: Python (NumPy, Pandas, Matplotlib), Java, C++, JavaScript, C, SQL, React.js, Node.js, Express.js, D/HTML, CSS, MERN Stack, Scikit-Learn, TensorFlow, Keras, OpenCV, CNNs, SVM, Logistic Regression, Random Forest, AutoEncoder Models, MongoDB, Git, GitHub

Soft Skills: Leadership & Team Management, Event Coordination & Public Speaking, Project Management, Analytical & Problem-Solving, Adaptability & Quick Learning, Independent and Collaborative, Result Oriented, Fluent in written and verbal communication in English and Hindi

Courses: Data Structures and Algorithms, Operating Systems, Database Management Systems, Software Engineering, Probability and Statistics, Machine Learning, Deep Learning, Computer Networks

LEADERSHIP & EXTRACURRICULARS

- Core Member of ACM-VIT Chapter and Mozilla Firefox Club
- Anchored and organized large-scale events, including a keynote session with YouTuber Love Babbar at Anna Auditorium
- Coordinated workshops, hackathons, and technical events with cross-team collaboration and administrative approvals

SUMMARY

I am a Computer Science student with experience in full-stack development and machine learning, seeking a Software Engineer Intern position. Eager to apply a strong foundation in object-oriented design and Java/C++ to contribute to meaningful projects and learn about agile development.